receptors (gustatory receptors), while the chloride ions activate amylase to digest starch. The IgA in the saliva acts against the microbes. The saliva helps in rolling the food into a ball so that it can be swallowed.

2.1.2.2 Gastric glands

Gastric juice is secreted by the different types of exocrine cells present in the gastric glands lining the stomach. These glandular cells empty their secretions into the gastric pit. There are three types of exocrine cells: Intrinsic factor and HCl secreting parietal cells, chief cells that secrete pepsinogen and gastric lipase, and mucous secreting mucosal cells. The secretions of the mucous, parietal, and chief cells form gastric juice, which amounts to 2 – 3 liters per day. The G cells, which are entero-endocrine (endocrine cells of the GI tract) in nature located mainly in the pyloric antrum secrete gastrin into the bloodstream.

2.1.2.3 Liver

The liver is the largest organ in the human body and it lies in the upper right section of the abdomen, just below the diaphragm. The liver has two main lobes, the bigger right lobe and the smaller left lobe, which are further divided into lobules that serve as the structural and functional units of liver (Figure 2.7). A lobule consists of hepatic cells arranged in groups that radiate from a central vein. Hepatic sinusoids separate these groups of cells from each other. Kupffer cells, which are phagocytic in nature are attached to the lining of the hepatic sinusoids (Figure 2.9).

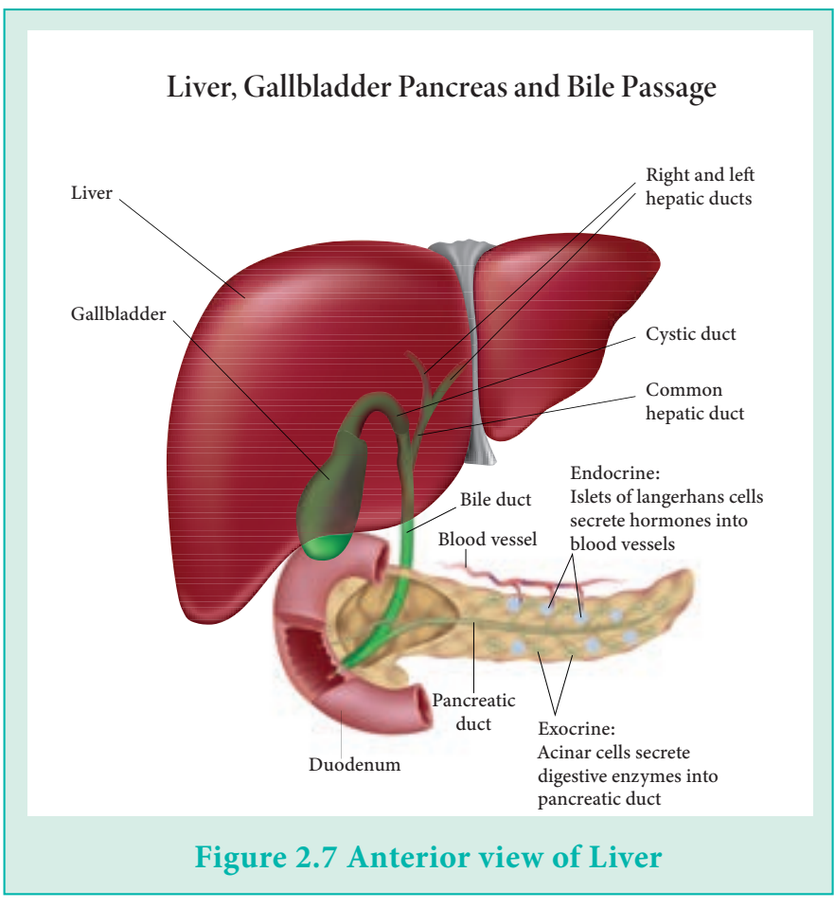


Figure 2.7 Anterior view of Liver

Every day, the hepatocytes secrete about 1 liter of bile, a yellow, olive-green or brown liquid with a pH of 7.6 – 8.6. Components of bile include bile salts, cholesterol, lecithin, bile pigments, and several ions.

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